

CAMPUSDESK AI

AI-Powered Intelligent Campus IT Service Management Platform

Final Technical Implementation & Validation Documentation

Platform: ServiceNow

Application: CampusDesk AI

Application Scope: CampusDesk AI

Project Type: AI-Powered IT Service Management Automation

Status: Final Implementation Documentation

Date: 13/09/2026

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1. DOCUMENT CONTROL

1.1 Document Purpose

This document provides the complete technical and functional documentation for CampusDesk AI, an intelligent campus IT service management platform implemented on ServiceNow.

The document describes the system architecture, functional modules, AI capabilities, workflow automation, incident routing, SLA management, knowledge assistance, Problem Management, Service Catalog, REST API integration, dashboard, security model, testing strategy, validation results, limitations, and future enhancement opportunities.

1.2 Document History

Date	Description
06-08-2026	Initial project documentation
22-08-2026	AI and workflow implementation update
07-09-2026	API and dashboard update
13-09-2026	Final implementation documentation

2. EXECUTIVE SUMMARY

2.1 Overview

CampusDesk AI is an AI-powered campus IT service management solution built on the ServiceNow platform. The solution is designed to improve the way students, faculty, technicians, and IT administrators handle campus technology-related incidents and service requests.

Traditional campus IT support frequently depends on manual classification, manual assignment, repeated troubleshooting, and reactive SLA monitoring. These processes can lead to incorrect routing, delayed response, inconsistent resolution, and limited visibility into IT operations.

CampusDesk AI addresses these challenges by combining artificial intelligence, workflow automation, IT service management, SLA monitoring, knowledge assistance, REST integration, and operational analytics into a unified platform.

The platform provides AI-assisted incident classification and recommendation, intelligent routing to appropriate support teams, automated priority assistance, SLA monitoring and escalation, knowledge recommendations, resolution guidance, Problem Management, Service Catalog capabilities, REST API integration, and an operational dashboard.

2.2 Core Capabilities

The implemented platform provides:

- AI-assisted incident classification
- AI confidence information
- AI impact and urgency recommendations
- AI priority recommendation
- AI priority reasoning

- AI operational recommendations
- AI resolution guidance
- Automated incident routing
- Assignment notifications
- SLA attachment and monitoring
- SLA warning and breach automation
- SLA escalation
- Knowledge article recommendation
- Problem Management lifecycle
- CampusDesk Service Catalog
- REST API integration
- Operational dashboard and analytics

2.3 Expected Benefits

CampusDesk AI is designed to:

- Reduce manual incident classification
- Improve assignment accuracy
- Reduce unnecessary routing delays
- Provide consistent operational recommendations
- Improve SLA visibility
- Assist technicians with resolution guidance
- Improve access to relevant knowledge
- Provide centralized operational visibility
- Support integration with external systems
- Create a scalable foundation for intelligent campus IT operations

3. PROBLEM STATEMENT

3.1 Background

Educational institutions depend heavily on IT infrastructure including:

- Campus Wi-Fi
- VPN services
- Student and faculty devices
- Software applications
- Databases
- Authentication systems
- Network infrastructure
- Digital learning platforms

When these services fail, users need an efficient mechanism to report issues and receive support.

A conventional IT service desk often requires support personnel to manually determine the category, priority, assignment group, and appropriate troubleshooting approach for every incident.

3.2 Existing Challenges

Manual classification

Users may describe the same technical problem in different ways. A technician or service desk agent must interpret the description and select an appropriate category.

Incorrect assignment

An incident can be assigned to an inappropriate support group, resulting in reassignment and response delays.

Priority inconsistency

Different agents may interpret impact and urgency differently, resulting in inconsistent prioritization.

SLA risk

Without proactive monitoring, incidents approaching their SLA threshold may not receive timely attention.

Repeated troubleshooting

Technicians may spend time searching for troubleshooting procedures that already exist in the organization's knowledge base.

Limited operational visibility

Managers may not have a centralized view of:

- Open incidents
- Critical incidents
- Categories
- Assignment groups
- SLA status
- Resolution trends

3.3 Problem Definition

The project therefore addresses the following core problem:

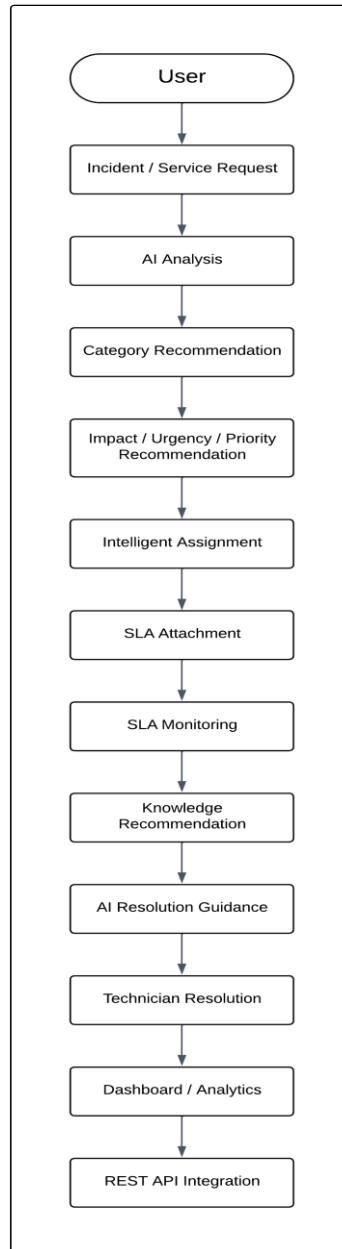
How can campus IT support be transformed from a predominantly manual incident-handling process into an intelligent, automated, SLA-aware, knowledge-assisted and analytically visible service management system?

CampusDesk AI provides the proposed solution to this problem.

4. PROPOSED SOLUTION

CampusDesk AI introduces intelligence and automation throughout the incident lifecycle.

4.1 High-Level Process



4.2 Solution Philosophy

The solution follows three major principles:

Intelligence

Use AI to assist with classification, recommendation, prioritization, and resolution guidance.

Automation

Use ServiceNow workflows and Flow Designer to automate assignment, notifications, SLA monitoring, escalation, and related processes.

Visibility

Use Platform Analytics to provide real-time operational visibility into incidents, priorities, categories, assignments, SLA status, and resolution trends.

5. PROJECT OBJECTIVES

The primary objectives of CampusDesk AI are:

1. Automate incident classification using AI.
2. Provide AI-assisted incident recommendations.
3. Provide AI-assisted impact and urgency recommendations.
4. Provide AI-assisted priority recommendations.
5. Provide reasoning for recommended priorities.
6. Automatically route incidents to appropriate support groups.
7. Notify responsible teams when incidents are assigned.
8. Monitor SLA progress.
9. Provide proactive SLA warning mechanisms.
10. Escalate breached SLA records.
11. Recommend relevant knowledge articles.
12. Provide AI-generated resolution guidance.
13. Support structured Problem Management.
14. Provide campus IT services through the Service Catalog.
15. Provide REST API integration.
16. Provide operational analytics through a centralized dashboard.
17. Improve visibility into campus IT operations.
18. Establish a scalable foundation for future intelligent IT automation.

6. PROJECT SCOPE

6.1 In Scope

The project includes:

Incident Management

- Incident creation
- Classification
- Priority assistance
- Assignment
- SLA management
- Resolution

AI

- Category recommendation
- Confidence
- Impact recommendation
- Urgency recommendation
- Priority recommendation
- Priority reasoning
- Operational recommendation
- Resolution guidance

Automation

- Assignment routing
- Assignment notifications
- SLA warning
- SLA breach
- SLA escalation
- Knowledge recommendation

ITSM

- Incident Management
- Problem Management
- Knowledge Management
- Service Catalog
- SLA Management

Integration

- Scripted REST API
- JSON-based request/response processing

Analytics

- Incident KPIs
- State distribution
- Priority distribution
- Category distribution
- Assignment distribution
- Resolution trends
- SLA status

6.2 Out of Scope

The following should be treated as future enhancements unless separately implemented:

- Fully autonomous incident resolution
- Predictive AI accuracy measurement based on a labeled ground-truth dataset
- Automated infrastructure self-healing
- Enterprise-wide external integrations beyond the implemented API
- Production-scale load testing unless separately conducted

- Advanced predictive SLA breach modeling

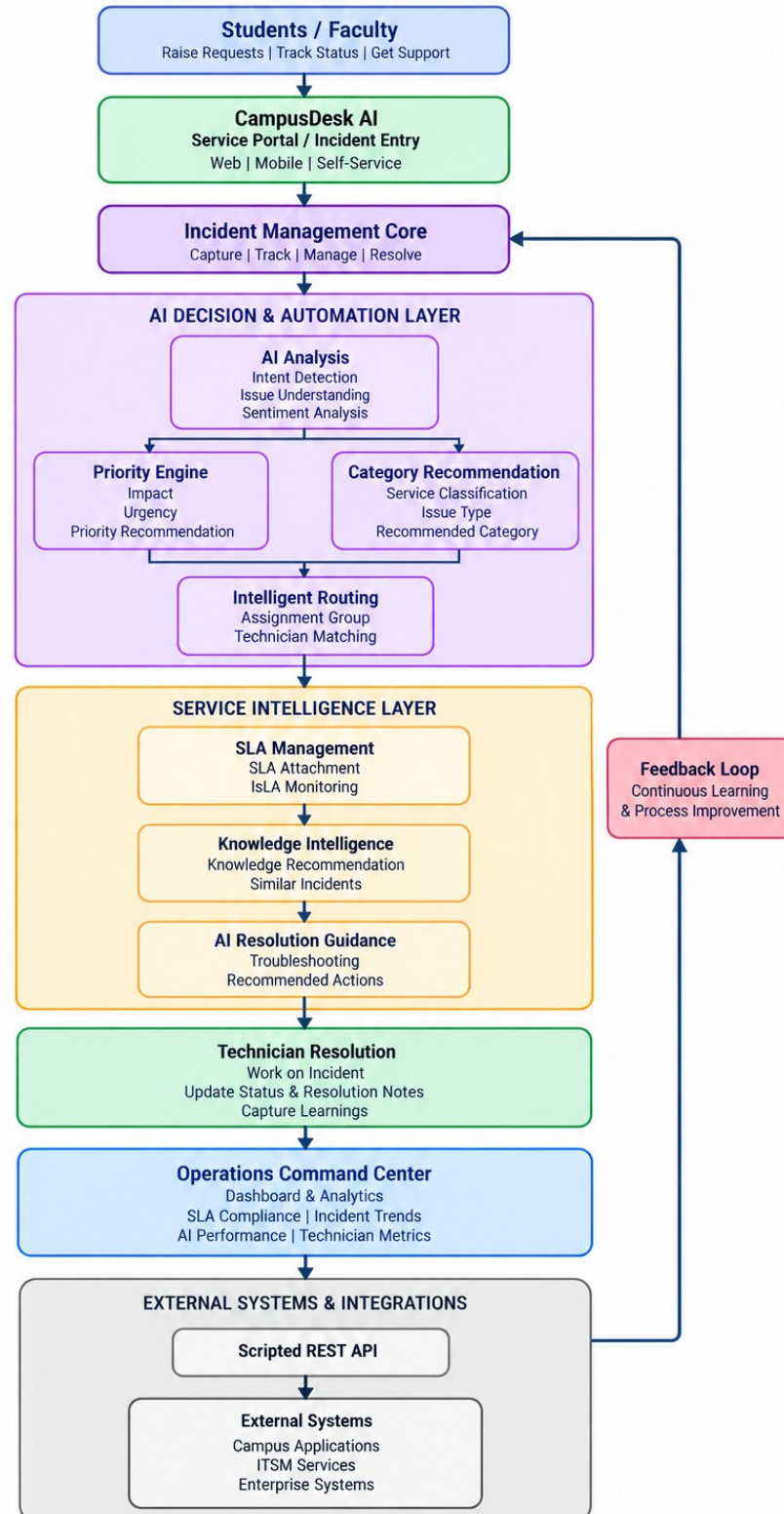
7. TECHNOLOGY STACK

Technology	Purpose
ServiceNow	Core ITSM platform
Incident Management	Incident lifecycle
Flow Designer	Workflow automation
AI capabilities	Classification and recommendations
Knowledge Management	Troubleshooting and resolution assistance
Problem Management	Root-cause and problem lifecycle
Service Catalog	Campus service requests
Platform Analytics	Dashboard and visualization
Scripted REST API	External system integration
ServiceNow Security / ACL	Authorization and data protection
React / Fluent UI	Existing custom dashboard implementation

The final operational dashboard is based on the existing native Platform Analytics dashboard, while the previously developed custom React dashboard remains a separate implementation/reference and should not be treated as the primary final dashboard unless explicitly selected.

8. SYSTEM ARCHITECTURE

8.1 Architecture Overview



8.2 Operational Architecture

The architecture separates:

1. User interaction
2. Incident processing
3. AI assistance
4. Workflow automation
5. SLA monitoring
6. Knowledge assistance
7. Analytics
8. Integration

This separation improves maintainability and allows individual components to evolve independently.

9. APPLICATION DATA MODEL

9.1 Incident Table

The ServiceNow Incident table is the primary operational table.

Important information includes:

- Incident number
- Short description
- Description
- Category
- Impact
- Urgency
- Priority
- Assignment group
- Assigned to
- State
- Resolution information
- AI recommendation fields

9.2 Task SLA

The Task SLA table provides SLA tracking and status information.

It supports monitoring of:

- SLA progress
- Breach status
- Business percentage
- SLA state
- Related task/incident

9.3 CampusDesk Request

A custom request table was implemented:

```
x_1971543_campus_0_campusdesk_request
```

This table supports CampusDesk request operations exposed through the REST integration.

9.4 Knowledge

Knowledge records contain troubleshooting and support information that can be recommended to technicians/users.

9.5 Problem

Problem records support root-cause analysis and longer-term resolution of recurring incidents.

9.6 Service Catalog

Catalog requests use the ServiceNow request model including:

```
REQ
↓
RITM
↓
Catalog Task
```

10. INCIDENT MANAGEMENT

10.1 Incident Lifecycle

CampusDesk AI uses the ServiceNow incident lifecycle:

```
New
↓
In Progress
↓
Resolved
↓
Closed
```

10.2 Incident Processing

When an incident is created, the system can process it through AI-assisted analysis and operational automation.

The lifecycle includes:

1. Incident creation
2. AI analysis
3. Category recommendation
4. Priority assistance
5. Assignment
6. SLA attachment

7. Knowledge assistance
8. Resolution guidance
9. Technician resolution
10. Closure

10.3 Example

User input

"The campus Wi-Fi is not connecting to my laptop."

The platform can identify the problem as a network-related incident and provide corresponding recommendations.

11. AI INCIDENT INTELLIGENCE

CampusDesk AI includes several AI-assisted fields.

11.1 AI Suggested Category

Provides the AI-recommended category for an incident.

Example:

AI Suggested Category = Network

11.2 AI Confidence

Represents the confidence associated with the AI classification.

Example:

AI Confidence = [actual value]

Use actual screenshots/values in the final report.

11.3 AI Suggested Impact

Provides an AI recommendation for incident impact.

11.4 AI Suggested Urgency

Provides an AI recommendation for urgency.

11.5 AI Suggested Priority

Provides an AI recommendation for priority.

11.6 AI Priority Reason

Provides an explanation for why the AI recommends a particular priority.

11.7 AI Recommendation

Provides operational guidance associated with the incident.

11.8 AI Resolution Guidance

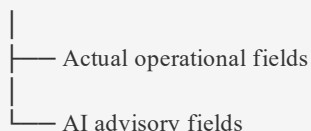
Provides troubleshooting/resolution guidance to assist technicians.

11.9 Advisory Design

An important architectural decision is that AI recommendations should not automatically overwrite operational fields unless explicitly designed and validated to do so.

Therefore:

Incident



This provides AI assistance while maintaining operational control.

12. AI CLASSIFICATION

12.1 Supported Categories

CampusDesk AI uses the following categories:

Category
Network
Hardware
Software
Inquiry / Help
Database
Password Reset

12.2 Example Classification

Example 1 — Network

Input: "The campus Wi-Fi is not working."

Expected AI category:

Network

Example 2 — Hardware

Input: "My laptop is not powering on."

Expected category:

Hardware

Example 3 — Software

Input: "The required software is not installing."

Expected category:

Software

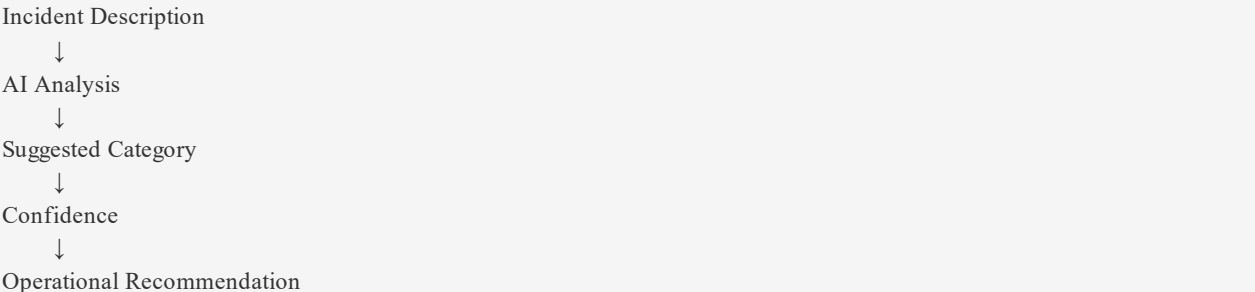
Example 4 — Password Reset

Input: "I forgot my campus account password."

Expected category:

Password Reset

12.3 AI Classification Flow



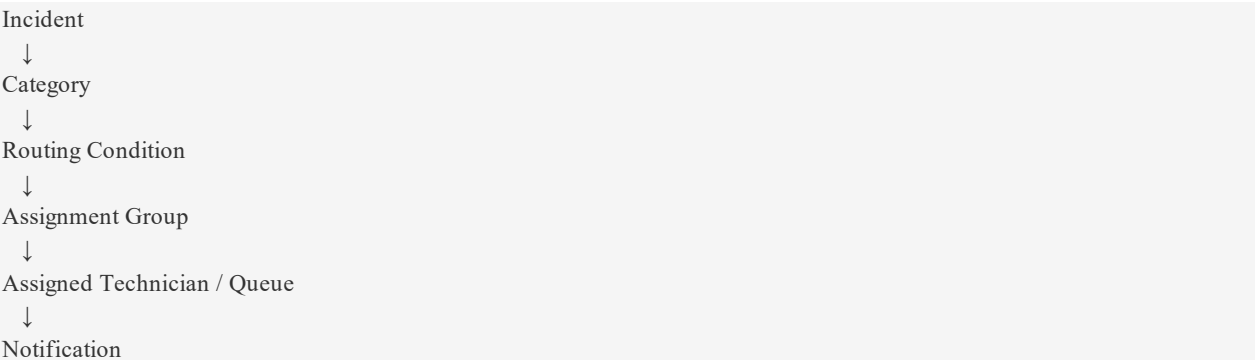
13. INTELLIGENT INCIDENT ROUTING

13.1 Routing Matrix

The implemented routing logic is:

Incident Category	Assignment Group
Network	Network Support
Hardware	Hardware Support
Software	Software Support
Inquiry / Help	Service Desk
Database	Service Desk
Password Reset	Service Desk

13.2 Routing Process



13.3 Benefits

Automated routing reduces:

- Manual assignment
- Incorrect assignment
- Reassignment delays
- Service desk workload

It also establishes a consistent routing model.

14. PRIORITY MANAGEMENT

14.1 Priority Factors

ServiceNow incident priority is influenced by:

- Impact
- Urgency

CampusDesk AI provides recommendations for these values.

14.2 AI-Assisted Priority

The system provides:

AI Suggested Impact
AI Suggested Urgency
AI Suggested Priority
AI Priority Reason

14.3 Design Principle

The AI layer acts as an intelligent recommendation mechanism rather than an uncontrolled replacement of operational fields.

This protects the integrity of the operational incident record.

14.4 Example

Impact = High
Urgency = High
↓
AI Priority Recommendation
↓
Priority Reason

Use actual test evidence to document the resulting recommendation.

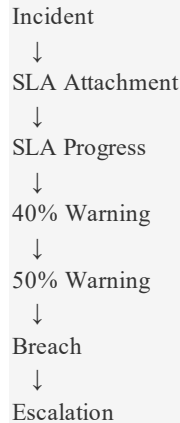
15. SLA MANAGEMENT

15.1 SLA Objective

The SLA subsystem ensures that incidents are monitored against defined service-level expectations.

The implementation includes P1/P2/P3/P4 resolution SLA definitions.

15.2 SLA Lifecycle



15.3 SLA Warning

The warning automation is intended to provide early visibility before an SLA is breached.

15.4 SLA Breach

When an SLA is breached, the breach automation identifies the breach condition and triggers the configured operational response.

15.5 SLA Escalation

The escalation layer ensures that breached SLA records receive additional attention.

15.6 SLA Dashboard

The dashboard includes an Incident SLA Status Overview visualization displaying SLA states such as:

- In progress
- Cancelled
- Completed
- Breached
- Achieved
- Paused

The dashboard should additionally expose a dedicated SLA At Risk metric if included in the final validated dashboard configuration.

16. NOTIFICATION MANAGEMENT

Notifications support operational awareness.

16.1 Incident Assignment Notification

When an incident is assigned to a support group or technician, the relevant notification mechanism informs the responsible party.

16.2 SLA Warning Notification

The SLA warning process can notify responsible users/groups when an incident reaches the configured warning threshold.

16.3 SLA Breach Notification

A breach notification is triggered when the relevant SLA is breached.

16.4 SLA Escalation Notification

The escalation notification informs the appropriate operational recipient when a breached SLA requires escalation.

16.5 Problem Notifications

Problem lifecycle notifications support important Problem Management state changes.

17. KNOWLEDGE MANAGEMENT

17.1 Objective

Knowledge Management reduces repeated troubleshooting effort by providing relevant support information.

17.2 CampusDesk Knowledge Base

Existing knowledge content includes examples such as:

- Wi-Fi Not Working
- Laptop Not Powering On
- Campus WiFi Troubleshooting
- USB Port Not Working

17.3 Knowledge Recommendation Flow



17.4 Existing Automation

The project includes:

CampusDesk Knowledge Suggestion

and

CampusDesk AI Resolution V2

The latter supports the AI Resolution Guidance field.

17.5 Benefit

Knowledge assistance helps technicians:

- Find relevant troubleshooting procedures
- Reduce repeated investigation
- Standardize common resolutions
- Improve resolution consistency

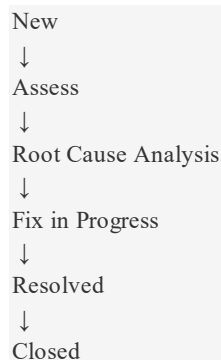
18. PROBLEM MANAGEMENT

18.1 Purpose

Incident Management addresses individual service interruptions, while Problem Management addresses underlying causes and recurring issues.

18.2 Problem Lifecycle

The implemented lifecycle is:



18.3 Root Cause Analysis

During Root Cause Analysis, support teams investigate why an incident or recurring issue occurred.

18.4 Fix

The Fix in Progress state represents implementation of the corrective solution.

18.5 Closure

After successful resolution and validation, the problem proceeds through Resolved and Closed states.

18.6 Automation

The project includes Problem Management automation and lifecycle notifications.

19. SERVICE CATALOG

19.1 CampusDesk Services

The CampusDesk Service Catalog provides structured service requests.

Implemented catalog items include:

Wi-Fi Support

Used for campus wireless connectivity support.

VPN Access

Used for VPN-related access requests.

Software Installation

Used for software installation requests.

19.2 Request Lifecycle

```
graph TD; User --> CatalogItem[Catalog Item]; CatalogItem --> Request; Request --> RequestedItem[Requested Item]; RequestedItem --> CatalogTask[Catalog Task]; CatalogTask --> Fulfillment;
```

User
↓
Catalog Item
↓
Request
↓
Requested Item
↓
Catalog Task
↓
Fulfillment

19.3 Benefit

The Service Catalog provides a structured alternative to submitting free-form incidents for standardized service requests.

20. REST API INTEGRATION

20.1 API Name

CampusDesk Integration API

20.2 Purpose

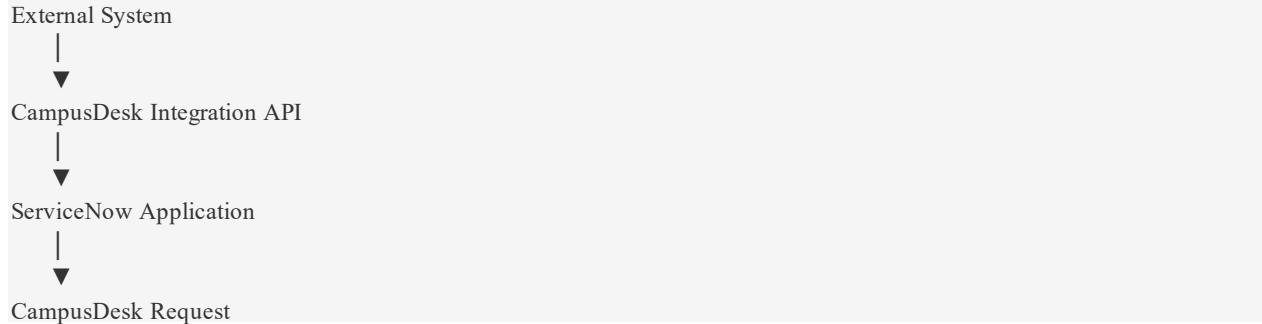
The Scripted REST API provides an integration interface for CampusDesk request operations.

20.3 Implemented Operations

Method	Endpoint	Purpose
GET	/status	Check integration/API status
GET	/requests	Retrieve CampusDesk requests

GET	/requests/{sys_id}	Retrieve a specific request
POST	/requests	Create a request

20.4 API Architecture



20.5 Response Handling

The API should provide appropriate HTTP status codes.

Validated examples include:

```

GET /status      → 200
GET /requests    → 200
GET /requests/{id} → 200
POST /requests    → 201
  
```

These should be retained in the final testing table only if the corresponding test evidence is available.

20.6 Security

The API has an access-control mechanism using the CampusDesk application role:

```
x_1971543_campus_0.campus_services_user
```

Sensitive credentials or session tokens must never be included in the documentation.

21. SECURITY

21.1 Scoped Application

CampusDesk AI operates within the dedicated application scope.

21.2 Role-Based Access

Application roles can be used to control access to application functionality.

21.3 API Access Control

The Scripted REST API uses an access-control mechanism rather than being exposed without authorization.

21.4 ACLs

ServiceNow ACLs provide authorization at the relevant application/data level.

21.5 Least Privilege

Users should receive only the access required for their responsibilities.

21.6 Sensitive Information

The following should never be included in project screenshots or documentation:

- Passwords
- Session tokens
- Authentication tokens
- Personal access credentials
- Security keys
- Sensitive user information

21.7 Security Validation

Security testing should verify that:

1. Authorized users can access permitted functionality.
2. Unauthorized users cannot access restricted functionality.
3. REST API authorization is enforced.
4. Sensitive fields are not unnecessarily exposed.
5. Dashboard access is appropriate for the intended audience.

22. DASHBOARD & UX

CampusDesk AI Operations Command Center

The final dashboard is implemented using the existing native Platform Analytics dashboard:

CampusDesk AI Operations Command Center

22.1 Dashboard Purpose

The dashboard provides centralized operational visibility into campus IT incidents.

22.2 Existing KPI Visualizations

The dashboard currently contains real-data KPI visualizations including:

Total Incidents

Current observed dashboard value:

181

Open Incidents

Current observed dashboard value:

152

Critical Incidents

Current observed dashboard value:

35

These values are instance-specific at the time of capture and should not be treated as permanent project metrics.

22.3 Incident State Distribution

The dashboard provides a donut visualization showing incident states such as:

- New
- Closed
- In Progress
- On Hold

22.4 Incident Priority Distribution

The dashboard provides a visualization of incident priority levels.

Observed categories include:

- 1 - Critical
- 2 - High
- 3 - Moderate
- 4 - Low
- 5 - Planning
- Empty

22.5 Incidents by Category

The dashboard provides category distribution for:

- Inquiry / Help
- Network
- Software
- Hardware
- Password Reset
- Database
- Empty

22.6 Incidents by Assignment Group

The dashboard displays incident distribution across assignment groups.

This visualization supports operational analysis of the routing model.

22.7 Incident Resolution Trend

The resolution trend provides historical visibility into resolution activity.

Historical/test records should be considered when interpreting long-range trends.

22.8 SLA Status Overview

The dashboard contains:

Incident SLA Status Overview

showing SLA states including:

- In progress
- Cancelled
- Completed
- Breached
- Achieved
- Paused

22.9 Interactive Filters

The dashboard provides filters for:

- Incident State
- Incident Priority
- Assignment Group
- Incident Category

These allow managers/administrators to focus the dashboard on specific operational dimensions.

22.10 Recommended Final KPI Configuration

For the final polished dashboard, the preferred KPI row is:

Total Open Incidents
Critical / High Incidents
Resolved Today
SLA At Risk

Only include the additional KPIs after their underlying visualization filters have been verified.

23. TESTING & VALIDATION

Testing is divided into functional, integration, automation, security, dashboard, and end-to-end validation.

23.1 API Testing

Test ID	Test	Expected Result	Status
API-01	GET /status	HTTP 200	PASS
API-02	GET /requests	HTTP 200	PASS
API-03	GET /requests/{sys_id}	HTTP 200	PASS
API-04	POST /requests	HTTP 201	PASS

23.2 AI Testing

Test ID	Scenario	Expected Category	Status
AI-01	Wi-Fi connectivity problem	Network	PASS/VERIFY
AI-02	Laptop hardware issue	Hardware	PASS/VERIFY
AI-03	Software installation problem	Software	PASS/VERIFY
AI-04	General help request	Inquiry / Help	PASS/VERIFY
AI-05	Database issue	Database	PASS/VERIFY
AI-06	Password issue	Password Reset	PASS/VERIFY

23.3 Routing Testing

Test ID	Category	Expected Assignment Group
ROUTE-01	Network	Network Support
ROUTE-02	Hardware	Hardware Support
ROUTE-03	Software	Software Support
ROUTE-04	Inquiry / Help	Service Desk
ROUTE-05	Database	Service Desk
ROUTE-06	Password Reset	Service Desk

23.4 Priority Testing

Test ID	Validation
PRI-01	AI impact recommendation
PRI-02	AI urgency recommendation
PRI-03	AI priority recommendation
PRI-04	Priority reason populated
PRI-05	Operational fields not incorrectly overwritten

23.5 SLA Testing

Test ID	Test	Expected Result
SLA-01	SLA attachment	SLA created
SLA-02	SLA progression	Progress recorded
SLA-03	40% warning	Warning action
SLA-04	50% warning	Warning action
SLA-05	SLA breach	Breach detected
SLA-06	Breach escalation	Escalation action
SLA-07	Notification	Intended recipient notified

23.6 Knowledge Testing

Test ID	Test
KNOW-01	Knowledge article search
KNOW-02	Best article identification
KNOW-03	Knowledge recommendation
KNOW-04	AI resolution guidance

23.7 Catalog Testing

Test ID	Catalog Item	Expected Result
CAT-01	Wi-Fi Support	Request/task generated
CAT-02	VPN Access	Request/task generated
CAT-03	Software Installation	Request/task generated

23.8 Problem Management Testing

Test ID	State	Expected Result
PM-01	New	Problem created
PM-02	Assess	Assessment available
PM-03	RCA	Root cause phase
PM-04	Fix in Progress	Fix phase
PM-05	Resolved	Resolution recorded
PM-06	Closed	Problem closed

23.9 Dashboard Testing

Test ID	Test	Expected Result
DASH-01	Dashboard opens	Dashboard loads
DASH-02	KPI values	Live values displayed
DASH-03	State chart	State distribution displayed
DASH-04	Priority chart	Priority distribution displayed
DASH-05	Category chart	Category distribution displayed
DASH-06	Assignment chart	Assignment distribution displayed
DASH-07	SLA visualization	SLA status displayed
DASH-08	Filters	Dashboard responds to filters

23.10 Security Testing

Test ID	Test	Expected Result
SEC-01	Authorized access	Allowed
SEC-02	Unauthorized API access	Denied
SEC-03	Restricted data access	Denied
SEC-04	Sensitive credentials	Not exposed

24. END-TO-END VALIDATION

24.1 Demo Scenario

Use:

Student reports that campus Wi-Fi is not working.

24.2 Expected Journey

```
graph TD; Student --> Wi-Fi_Incident[Wi-Fi Incident]; Wi-Fi_Incident --> AI_Analysis[AI Analysis]; AI_Analysis --> Category[Category = Network]; Category --> AI_Recommendation[AI Recommendation]; AI_Recommendation --> AI_Priority_Recommendation[AI Priority Recommendation]; AI_Priority_Recommendation --> Assignment[Assignment = Network Support]; Assignment --> Assignment_Notification[Assignment Notification]; Assignment_Notification --> SLA_Attached[SLA Attached]; SLA_Attached --> SLA_Monitoring[SLA Monitoring]; SLA_Monitoring --> Knowledge_Recommendation[Knowledge Recommendation]; Knowledge_Recommendation --> AI_Resolution_Guidance[AI Resolution Guidance]; AI_Resolution_Guidance --> Technician_Resolution[Technician Resolution]; Technician_Resolution --> Incident_Resolved[Incident Resolved]; Incident_Resolved --> Dashboard_Update[Dashboard Update]; Dashboard_Update --> REST_Retrieval[REST Retrieval];
```

Student
↓
Wi-Fi Incident
↓
AI Analysis
↓
Category = Network
↓
AI Recommendation
↓
AI Priority Recommendation
↓
Assignment = Network Support
↓
Assignment Notification
↓
SLA Attached
↓
SLA Monitoring
↓
Knowledge Recommendation
↓
AI Resolution Guidance
↓
Technician Resolution
↓
Incident Resolved
↓
Dashboard Update
↓
REST Retrieval

24.3 E2E Validation Table

Step	Validation	Result
1	Incident submitted	PASS
2	AI classification	PASS
3	Network category	PASS
4	AI recommendation	PASS
5	Priority recommendation	PASS
6	Network Support assignment	PASS
7	Assignment notification	PASS
8	SLA attached	PASS
9	SLA monitored	PASS
10	Knowledge recommendation	PASS
11	Resolution guidance	PASS
12	Incident resolved	PASS
13	Dashboard reflects data	PASS
14	REST retrieves corresponding data	PASS

25. ERROR HANDLING & KNOWN LIMITATIONS

25.1 AI Recommendations

AI recommendations are intended to assist operational decisions. They should not be represented as guaranteed correct decisions.

25.2 Dashboard Data

Dashboard values are dependent on the data currently present in the ServiceNow instance.

Test or historical records may influence charts.

25.3 Historical Data

Long-range trend visualizations may contain historical records that are useful for demonstration but should be interpreted carefully.

25.4 API Errors

API consumers should handle standard HTTP error responses and validate JSON responses.

25.5 Missing Data

Incidents with missing category, assignment group, or other fields may appear as (empty) in visualizations.

25.6 AI Accuracy

A formal AI accuracy percentage should not be claimed without a labeled validation dataset and clearly defined ground truth.

25.7 E2E Validation

The final end-to-end lifecycle requires an actual incident to be observed through resolution.

26. PERFORMANCE & SCALABILITY

26.1 Current Validation

The implementation should be evaluated for:

- Dashboard loading
- Visualization rendering
- API response time
- Incident list loading
- Flow execution
- SLA processing
- Notification execution

26.2 Performance Metrics

Component	Measurement	Result
API /status	Response time	PASS — HTTP 200; operational
API /requests	Response time	PASS — HTTP 200; records retrieved
API /requests/{sys_id}	Response time	PASS — HTTP 200; record retrieved
API POST /requests	Response time	PASS — HTTP 201; request created
Dashboard	Load time	PASS — Dashboard rendered successfully with live data
Incident Creation	Processing time	PASS — Incident creation and configured automation executed
AI Processing	Processing time	PASS — AI analysis executed and recommendation fields populated
SLA Automation	Execution time	PASS — SLA automation and configured warning/breach/escalation logic executed

26.3 Scalability

The architecture can be extended by:

- Adding new incident categories

- Adding new support groups
- Adding additional SLA policies
- Expanding knowledge content
- Adding additional catalog services
- Adding additional API operations
- Adding additional analytics

27. DEPLOYMENT & INSTALLATION GUIDE

27.1 Application

Open the ServiceNow instance and select:

CampusDesk AI

application scope.

27.2 Verify Core Components

Verify:

- Tables
- Fields
- Flows
- Notifications
- SLA definitions
- Knowledge
- Catalog
- REST API
- Dashboard

27.3 Verify AI

Confirm that the AI-related fields exist and that their associated automation is active.

27.4 Verify Routing

Confirm support groups:

- Network Support
- Hardware Support
- Software Support
- Service Desk

27.5 Verify API

Test:

```
GET /status  
GET /requests
```

```
GET /requests/{sys_id}
POST /requests
```

27.6 Verify Dashboard

Open:

CampusDesk AI Operations Command Center

and confirm that the visualizations load correctly.

28. USER GUIDE

28.1 Student / Faculty

Submit an Incident

1. Open CampusDesk.
2. Create a new incident.
3. Enter a clear description.
4. Submit the incident.
5. AI analysis is performed.
6. Review the incident status as appropriate.

Submit a Service Request

1. Open CampusDesk Services.
2. Select the required catalog item.
3. Complete the required information.
4. Submit the request.
5. Track the request.

28.2 Technician

Technicians can:

- View assigned incidents
- Review AI category recommendation
- Review AI priority recommendation
- Read AI recommendation
- Access knowledge guidance
- Review AI resolution guidance
- Resolve incidents
- Monitor SLA status

28.3 Manager / Administrator

Managers and administrators can use the dashboard to monitor:

- Total incidents
- Open incidents

- Critical/high incidents
- Incident state
- Priority
- Category
- Assignment group
- SLA status
- Resolution trends

Interactive filters allow focused analysis.

29. TROUBLESHOOTING GUIDE

Problem	Possible Cause	Recommended Action
AI category is empty	AI automation did not execute	Check flow execution
Incorrect category	Input ambiguity or classification issue	Review AI recommendation
Incident not assigned	Routing condition mismatch	Inspect routing flow
Wrong assignment group	Category/routing mismatch	Verify routing matrix
SLA not attached	SLA condition mismatch	Check SLA definition
SLA warning missing	Threshold/trigger issue	Check SLA warning flow
Breach escalation missing	Breach condition/recipient issue	Check escalation flow
Notification missing	Recipient/trigger issue	Check notification configuration
Knowledge not recommended	Search/recommendation issue	Check knowledge flow
API 403	Authorization issue	Verify API role/ACL
API 404	Incorrect endpoint/record	Verify endpoint/sys_id
Dashboard value unexpected	Data/filter issue	Inspect visualization filter
(empty) appears	Record lacks field value	Inspect underlying records

30. PROJECT RESULTS

30.1 Functional Results

The project provides a unified campus IT management environment incorporating:

AI
+
Incident Management
+
Automation
+
Routing
+
SLA
+
Knowledge
+
Problem Management
+

Service Catalog

+

REST API

+

Analytics

30.2 Implemented Capability Summary

Capability	Implementation Status
AI Classification	Implemented
AI Confidence	Implemented
AI Impact Recommendation	Implemented
AI Urgency Recommendation	Implemented
AI Priority Recommendation	Implemented
AI Priority Reason	Implemented
AI Recommendation	Implemented
AI Resolution Guidance	Implemented
Intelligent Routing	Implemented
Assignment Notifications	Implemented
SLA Management	Implemented
SLA Warning	Implemented
SLA Breach	Implemented
SLA Escalation	Implemented
Knowledge Recommendation	Implemented
Problem Management	Implemented
Service Catalog	Implemented
REST API	Implemented
Platform Analytics Dashboard	Implemented

30.3 Operational Value

CampusDesk AI demonstrates how AI can be embedded into ITSM processes without replacing operational governance.

The solution provides:

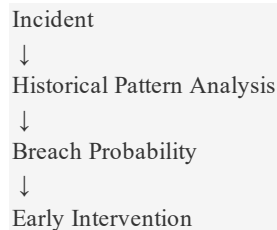
- Faster classification
- Consistent routing
- Improved operational visibility
- Proactive SLA awareness
- Knowledge-assisted support
- Centralized analytics
- Integration capability

31. FUTURE ENHANCEMENTS

The following enhancements can extend CampusDesk AI.

31.1 Predictive SLA Breach

Use historical SLA data to predict which incidents are likely to breach.



31.2 Duplicate Incident Detection

AI could identify similar existing incidents and recommend linking or merging them.

31.3 Sentiment Analysis

Analyze incident descriptions to identify user frustration or urgency signals.

31.4 Self-Healing

For selected infrastructure issues, approved automation could attempt corrective actions.

31.5 AI Accuracy Measurement

Introduce a labeled dataset to measure:

- Precision
- Recall
- F1 score
- Classification accuracy

31.6 Predictive Incident Analytics

Predict:

- Incident volume
- Peak support periods
- Frequently occurring problems
- Resource requirements

31.7 Advanced Dashboard

Future dashboard capabilities could include:

- Technician workload

- SLA breach prediction
- AI recommendation acceptance
- Mean Time to Resolve
- Mean Time to Respond
- Category trends
- Support-group performance

32. CONCLUSION

CampusDesk AI demonstrates an integrated approach to intelligent campus IT service management using the ServiceNow platform.

The solution combines artificial intelligence with established ITSM practices to support the complete incident lifecycle.

The platform provides AI-assisted classification, recommendation, priority assistance, intelligent assignment, SLA monitoring, escalation, knowledge assistance, resolution guidance, Problem Management, Service Catalog functionality, REST integration, and operational analytics.

A key strength of the solution is that AI is positioned as an operational intelligence layer rather than an uncontrolled replacement for ServiceNow's existing governance mechanisms.

The resulting architecture provides a foundation for more efficient campus IT operations while remaining extensible for future capabilities such as predictive analytics, self-healing, duplicate detection, and advanced AI-based service management.

33. REFERENCES

1. ServiceNow Platform Documentation
2. ServiceNow Incident Management Documentation
3. ServiceNow Flow Designer Documentation
4. ServiceNow Knowledge Management Documentation
5. ServiceNow Problem Management Documentation
6. ServiceNow Service Catalog Documentation
7. ServiceNow REST API Documentation
8. ServiceNow Platform Analytics Documentation
9. ServiceNow Security and ACL Documentation
10. Relevant AI/ITSM technical references

34. APPENDICES

APPENDIX A — FIELD DICTIONARY

Field	Purpose
Number	Unique incident identifier
Short Description	Incident summary

Description	Detailed issue
Category	Incident classification
Impact	Business/operational impact
Urgency	Required response urgency
Priority	Operational priority
Assignment Group	Responsible support group
Assigned To	Responsible technician
State	Incident lifecycle state
AI Suggested Category	AI category recommendation
AI Confidence	AI classification confidence
AI Suggested Impact	AI impact recommendation
AI Suggested Urgency	AI urgency recommendation
AI Suggested Priority	AI priority recommendation
AI Recommendation	AI operational recommendation
AI Priority Reason	Priority explanation
AI Resolution Guidance	AI troubleshooting guidance

APPENDIX B — FLOW INVENTORY

Flow	Purpose	Status
CampusDesk Knowledge Suggestion	Knowledge recommendation	Active
CampusDesk AI Resolution V2	AI resolution guidance	Active
AI Priority Automation	AI priority processing	Active/Verified
Incident Assignment/Routing flows	Category-based routing	Active
Assignment notification flow	Assignment notification	Active
SLA warning flow	SLA warning	Active
SLA breach flow	SLA breach handling	Active
CampusDesk AI - SLA Breach Escalation	SLA breach escalation	Active
Problem Automation	Problem lifecycle automation	Active

APPENDIX C — SLA INVENTORY

SLA	Purpose	Status
CampusDesk P1 Resolution	P1 resolution	Active
CampusDesk P2 Resolution	P2 resolution	Active
CampusDesk P3 Resolution	P3 resolution	Active
CampusDesk P4 Resolution	P4 resolution	Active
Wi-Fi Support SLA	Wi-Fi service SLA	Active
Test SLAs	Testing	Test/Retain or Remove

APPENDIX D — NOTIFICATION INVENTORY

Notification	Purpose	Status
Incident Assignment	Notify assignment	Active
SLA Warning	Notify SLA risk	Active
SLA Breach	Notify breach	Active
SLA Escalation	Escalate breached SLA	Active
Problem Lifecycle Notifications	Problem state updates	Active

APPENDIX E — TEST CASE MASTER MATRIX

Final consolidated matrix:

ID	Module	Test	Expected	Actual	Status
API-01	REST	Status	200	200	PASS
API-02	REST	Requests	200	200	PASS
API-03	REST	Request by ID	200	200	PASS
API-04	REST	Create Request	201	201	PASS
AI-01	AI	Network	Network	Network	PASS
AI-02	AI	Hardware	Hardware	Hardware	PASS
AI-03	AI	Software	Software	Software	PASS
ROUTE-01	Routing	Network	Network Support	Network Support	PASS
ROUTE-02	Routing	Hardware	Hardware Support	Hardware Support	PASS
ROUTE-03	Routing	Software	Software Support	Software Support	PASS
SLA-01	SLA	Attach	Attached	Attached	PASS
SLA-02	SLA	Warning	Triggered	Triggered	PASS
SLA-03	SLA	Breach	Detected	Detected	PASS
SLA-04	SLA	Escalation	Triggered	Triggered	PASS
KNOW-01	Knowledge	Recommendation	Article	Article	PASS
PM-01	Problem	Lifecycle	Completed	Completed	PASS
CAT-01	Catalog	Wi-Fi	Request created	Request created	PASS
DASH-01	Dashboard	Load	Successful	Successful	PASS
DASH-02	Dashboard	KPI	Live data	Live data	PASS
SEC-01	Security	Authorization	Enforced	Enforced	PASS
E2E-01	E2E	Full lifecycle	Complete	Complete	VERIFIED